

Scientific Method in Data Science

Data Science: Theory, Practice and Societal Implications

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Part I – Introduction



1. What will we learn today?

- Explain how the scientific method applies to data-science projects.
- Turn a broad problem into a measurable research question.
- Distinguish exploratory, descriptive, predictive, and causal questions.
- Formulate testable hypotheses and identify relevant variables.
- Evaluate data quality, bias, and fitness for purpose.
- How to integrate machine learning in the scientific method.

2. Is data science a science?

- Let's start with a familiar decision: *Choosing a restaurant from ratings.*
 - Where evidence, assumptions, and judgment enter the decision?



- For being rigorous we could
 1. Explore the data: location, options, budget, reviews...
 2. Building the decision model: rule-based model using most relevant features
 3. Interpreting the results: everyone agrees on the chosen place?
 4. Make a decision: book the place

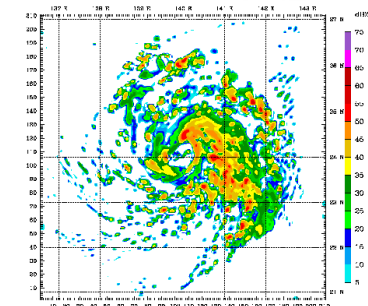
3. Four approaches to knowledge

- **Empirical Observation:** Gathering facts by directly watching, measuring, and testing the physical world.
 - **How it works:** Scientists run physical experiments and collect raw data from nature.
- **Theory and Models:** Using logic, mathematics, and reasoning to explain observations and predict future events.
 - **How it works:** Scientists create abstract rules and formulas that describe how systems behave.
- **Computational Simulation:** Using computers to model and run experiments on complex systems that are too difficult or dangerous to test in real life.
 - **How it works:** Software calculates thousands of variables over time to show what might happen.
- **Data-Intensive Scientific Discovery:** Using massive data sets, algorithms, and artificial intelligence to find new patterns without starting from a traditional hypothesis.
 - **How it works:** Computers scan petabytes of information to spot hidden connections that human eyes miss.



$$F = G \frac{m_1 m_2}{r^2}$$

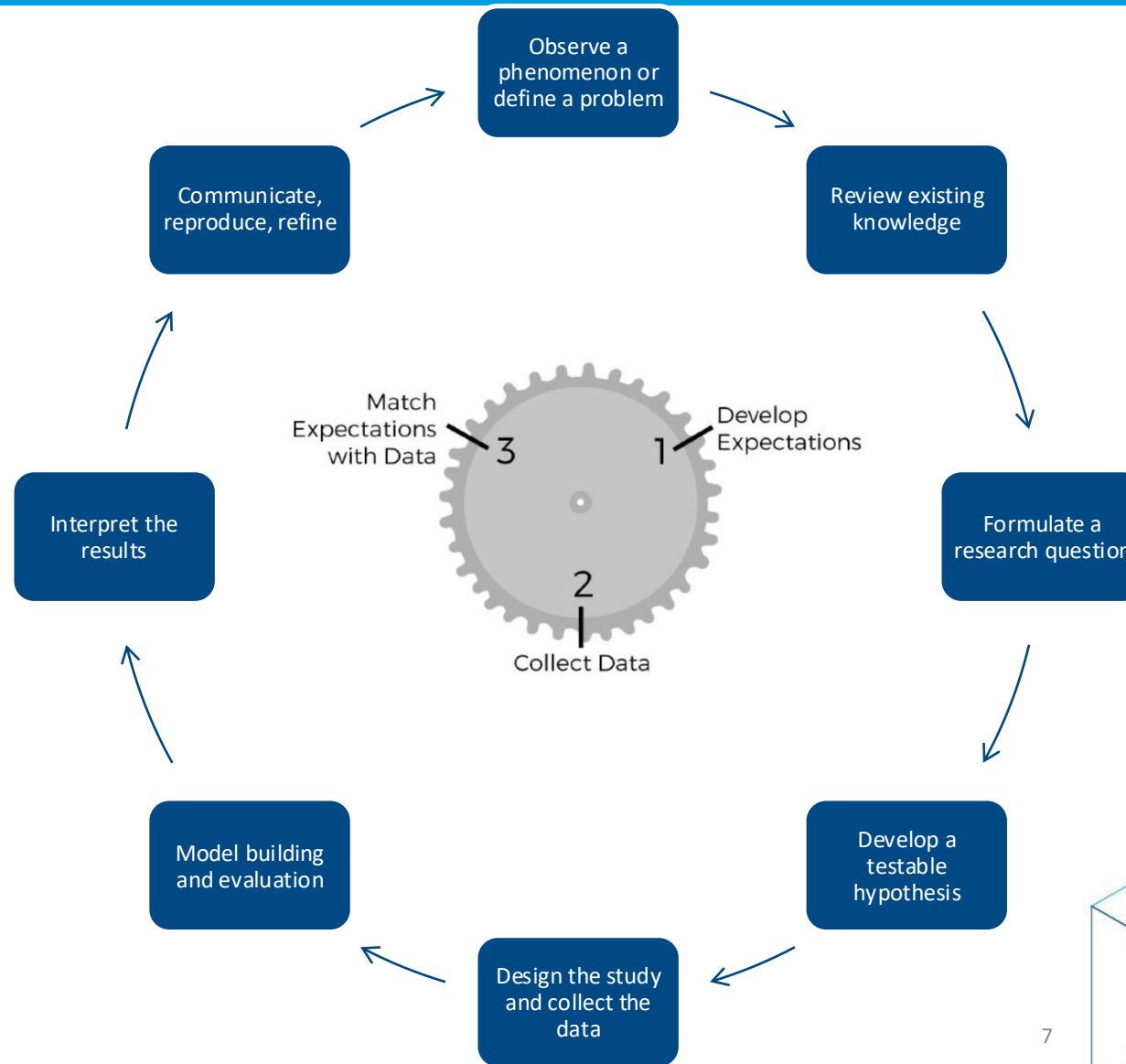
Labels: force (F), Gravitational constant (G), mass (m₁, m₂), distance (r).



Part II – The scientific method



4. The scientific method as an iterative cycle



5. From a real-world problem to a data question

Real world problem	Measurable research question
"Students are struggling academically"	"Among first-year students, what is the relationship between weekly study time and final course grades?"

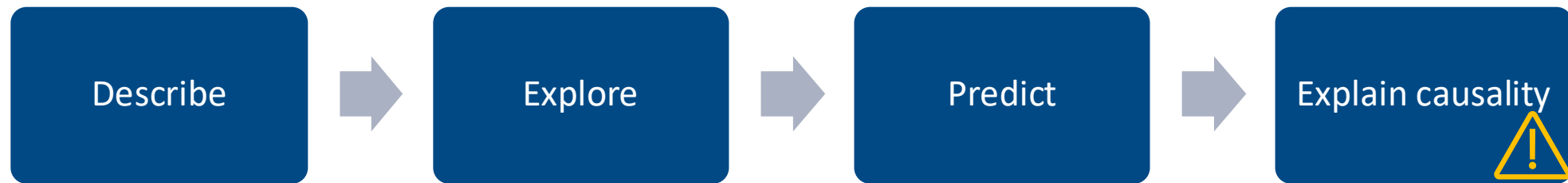
Operationalization: Translating an abstract concept into something that can be observed or measured

A useful data question defines:

- Population: First-year students
- Unit of analysis: One student
- Predictor: Weekly study time
- Outcome: Final course grade
- Time frame: One academic term
- Relationship of interest: Association between study time and grades

4. From a real-world problem to a data question

Different operational definitions can lead to different findings



- **Descriptive:** How many hours do students study each week?
- **Exploratory:** What study-behavior patterns appear in the data?
- **Predictive:** Can study behavior predict a student's final grade?
- **Causal:** Would increasing study time improve final grades?

The question determines the data, method, and conclusions—not the other way around.

5. Hypothesis, evidence, and study design

Research question

Is weekly study time associated with final course grade among first-year students?

Hypotheses

- **Null hypothesis (H_0):** Weekly study time is not associated with final grades
- **Alternative hypothesis (H_1):** Weekly study time is associated with final grades

Variables

- **Predictor:** Weekly study time
- **Outcome:** Final course grade
- **Possible confounders:** Prior achievement, attendance, motivation, employment, course difficulty...

Study design

- **Observational study:** Measure existing study habits
- **Controlled experiment:** Assign different study routines

Evidence

- **Observational:** Study time is associated with grades
- **Experimental:** The intervention caused an estimated change in grades

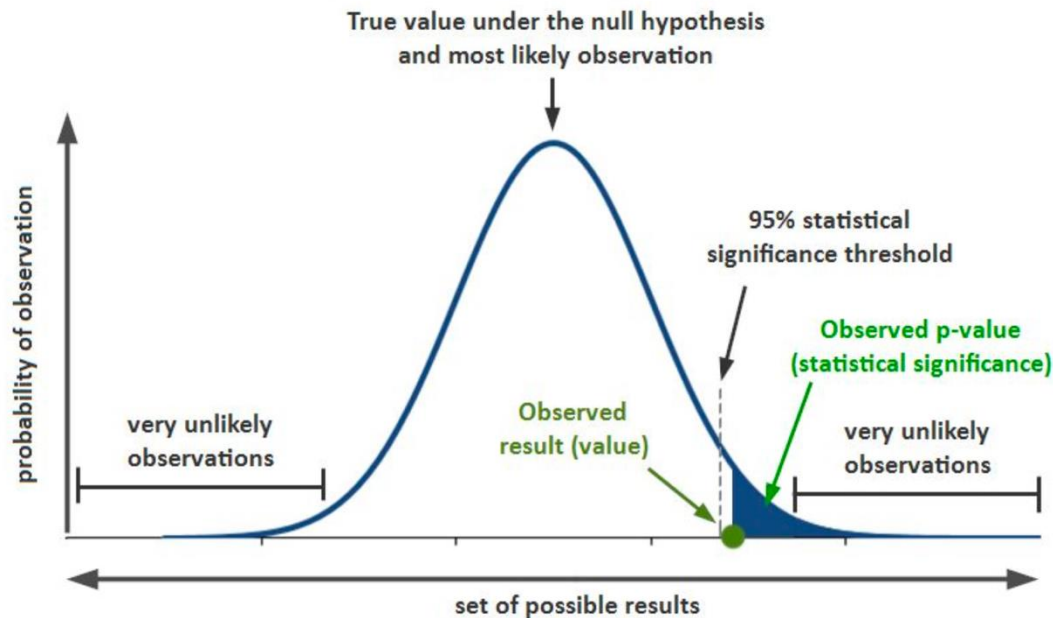
5. Hypothesis, evidence, and study design

- A **hypothesis** converts a research question into a claim that can be evaluated using evidence. It must be:
 - Specific
 - Measurable
 - Testable
 - Falsifiable
- Null hypothesis (H_0): represents no effect, no difference, or no association.
- Alternative hypothesis (H_1): represents the effect or relationship being investigated.
- Hypothesis **are not proved with certainty**. Instead, it is evaluated whether the observed evidence is reasonable compatible with the null hypothesis, reporting the uncertainty surrounding their estimates

P-value: “If the null model and its assumptions were correct, how unusual would evidence like this be?”

5. Hypothesis, evidence, and study design

Probability & Statistical Significance Explained



- The p-value is a statistical number that shows the change of getting this result if there is no real effect or difference happening.
- A small p-value may indicate tension between the data and the null model, but *interpretation* should also consider:
 - Effect size
 - Confidence intervals
 - Sample size
 - Study design
 - Data quality
 - Practical importance

5. Hypothesis, evidence, and study design

Main paradigms for the study design:

Observational approach	Experimental approach
<i>Measure existing study habits</i> • Usually more practical • Identifies associations • Vulnerable to confounding	<i>Assign or encourage different study routines</i> • Can provide stronger causal evidence • May help estimate causal effects • Requires ethical and practical safeguards
Evidence: “Study time is associated with grades.” However, other explanations remain possible: • Highly motivated students may both study less and perform better. • Students with strong prior knowledge may need less study time. • Students working long hours may have less time to study. • Students struggling with a subject may study more but still receive lower grades.	Evidence: “The intervention caused an estimated change in grades.” However, experiments also present challenges: • Students cannot ethically be prevented from studying • Participants may not follow their assigned routine. • A short intervention may not represent normal behavior. • Results from volunteers may not generalize to all students.

Part III – Coming back to the Data



6. Data as scientific evidence



<i>Is weekly study time associated with final grades?</i>	
Possible measure	Potential limitation
Student survey	Recall and social-desirability bias
Learning-platform activity	Misses studying outside the platform
Library entry records	Presence does not necessarily mean studying
Study-tracking application	May change behavior and invade privacy

6. Data as scientific evidence

Not all available data are suitable evidence. Before analyzing a dataset, ask:

- **Relevant:** Does it measure the concepts in the research question?
- **Accurate:** Are the values correct and measured consistently?
 - Potential problems: Incorrect values, duplicate records, inconsistent units, changes in definitions, different data-collection procedures...
- **Complete:** Are important observations or variables missing?
 - E.g. Participants skip sensitive questions, a device or sensor fails, records are removed during data cleaning.
- **Current:** Does the time period fit the decision?
- **Representative:** Does the dataset reflect the target population?
 - A large dataset may not be representative if **samples** do not belong to the **target population**.
- **Ethical:** Was the data collected and used responsibly?
 - Anonymization reduces some risks but does not automatically make every use ethical.

7. Qualitative, Quantitative, and Mixed Research

- **Qualitative research:** Explores experiences, meanings, motivations, and context.
 - Data: interviews, observations, documents, open-ended responses
 - Questions: “How?” and “Why?”
 - Analysis: coding, themes, narratives, sentiment
- **Quantitative research:** Measures variables, differences, relationships, and effects.
 - Data: numerical measurements, categories, surveys, experiments
 - Questions: “How much?”, “How often?”, “Is there a relationship?”
 - Analysis: descriptive statistics, hypothesis tests, regression, prediction
- **Mixed methods:** Combines both to understand the pattern and its explanation.

In the study-time example:

- Quantitative: Is weekly study time associated with final grades?
- Qualitative: How do students describe the barriers that affect their study habits?
- Mixed-methods conclusion: Quantitative data reveal the pattern; qualitative data help explain why it occurs.

7. Qualitative, Quantitative, and Mixed Research

Qualitative	Quantitative
More than one form of data collected in each study.	Numerical data
Data in forms of words: open-ended questions (interviews), documents, subjective observations	Data in form of measures: continuous values or defined categories
Workflow: Initial data reduction & open coding → Category building → Thematic mapping & exploration → Synthesis & reporting	Workflow: Data cleaning & structuring → Univariate analysis → Bivariate & multivariate analysis → Insights & Feature selection

- Angry
- Sleepy
- Long, untidy fur
- Different colors



- Weight: 4kg
- Gender: female
- Age: 6 years
- Height: 30cm
- Jumps up to 2m

7. Qualitative, Quantitative, and Mixed Research

Qualitative	Quantitative
Advantages: • Useful during the early stages of a study if the researcher is unsure of what to focus on. • Does not need a strict design plan before it begins, which gives the freedom to let the study unfold more naturally. • Gains more detailed and rich data in the form of comprehensive written descriptions or visual evidence. • Looks at context and social meaning, which is advantageous particularly in the social sciences	Advantages: • Allows the researcher to measure and analyze data. • Relationship between an independent and dependent variable is studied in detail, so the researcher is more objective about the findings. • Used to test hypotheses in experiments because of its ability to measure data using statistics
Disadvantages: • Researcher is heavily involved in the process, which gives them a subjective view of the study and its participants. • Researcher interprets the research according to their own biased view, which skews the data gathered. • Very time-consuming and can last for months or even years.	Disadvantages: • The context of the study or experiment may be ignored. • Does not discuss the meaning things have for different people as qualitative research does. • In some cases, a large sample of the population must be studied; to obtain more statistically accurate the results can be. As such, the timeline, complexity, and cost of the research will be increased.

8. More data versus better evidence

- **More data** increase the number of observations.
- **Better evidence** comes from appropriate measurements, representative sampling, credible design, and transparent analysis.

A very large biased dataset can produce a precise estimate of the wrong quantity

Problem: A university wants to understand student engagement. It uses the number of clicks in its online learning platform as the only measure.

1. What concept is the university trying to measure?
2. Is the number of clicks a valid measure of engagement?
3. Which students or activities might be overlooked?
4. What biases could enter the dataset?
5. What additional quantitative or qualitative data would improve the evidence?

Part IV – Analysis, Interpretation, and Common Traps



9. Finding a pattern is only the beginning

- Three types of claims:
 1. Association: variables move together
 2. Prediction: one variable helps forecast another
 3. Causation: changing one variable changes another
 - It requires the strongest study design!

<i>Students who study the most tend to receive higher grades</i>	
Association claim:	Study time and grades vary together.
Prediction claim:	Study-related variables help predict grades.
Causation claim:	Changing study time produces a change in grades.



Our study may indicate an association, but it does not immediately tell us:

- Whether studying caused the higher grades
- Whether the relationship applies to all students
- Whether another variable explains the pattern
- Whether the effect is large enough to matter
- Whether the pattern will appear in new data

Do not make a stronger claim than the data and study design support

10. Exploratory versus confirmatory analysis

- **Exploratory data analysis** is used to discover patterns, unusual observations, and possible relationships.
 - Examples include:
 - Inspecting distributions
 - Comparing groups
 - Creating visualizations
 - Looking for clusters
 - Identifying unexpected relationships
- **Confirmatory analysis** evaluates a question or hypothesis specified before examining the results. Ideally, the researcher defines in advance:
 - The hypothesis
 - The variables
 - The sample
 - The analytical method
 - The criteria for interpreting the result
- A useful principle: Explore on one dataset; confirm on another.

Exploration is essential, but patterns discovered in a dataset should not automatically be treated as confirmed findings

11. Association is not causation

Two variables may be associated because:

- The first affects the second.
- The second affects the first.
- A third variable affects both.
- The relationship results from selection, measurement, or chance.

Common analytical traps:

- Spurious correlation
- Simpson's paradox
- Multiple testing
- Overfitting
- Data leakage
- Statistical significance without practical importance



12. Analytical traps

Spurious correlation

- A spurious correlation is a statistical relationship that does not reflect a meaningful causal connection.
- In data science, spurious relationships can still produce apparently accurate models, particularly when the model learns shortcuts that do not generalize.



Source: <https://tylervigen.com>

12. Analytical traps

Simpson's paradox

- Simpson's paradox occurs when a relationship observed in aggregated data changes or reverses after the data are divided into relevant groups.

Students who study the most tend to receive higher grades

Across all students, more study time may appear to be associated with lower grades. However, students in difficult courses may both study more and receive lower grades than students in easier courses.

Which important groups or contextual variables might be hidden by an overall average?

12. Analytical traps

Multiple testing and data dredging

- If researchers test enough variables, some relationships will appear statistically significant by chance.
- Good practice include:
 - Defining hypotheses before testing
 - Limiting unnecessary comparisons
 - Correcting for multiple tests
 - Validating findings with new data
 - Reporting all relevant analyses, not only successful ones

Students who study the most tend to receive higher grades

Testing grades against dozens of behaviors, demographic variables, and platform activities may produce several small p-values even when no meaningful relationships exist.

12. Analytical traps

Overfitting

- An overfitted model learns noise and peculiarities in the training data instead of a generalizable pattern.
- Signs include:
 - Excellent performance on training data
 - Much weaker performance on unseen data
 - Excessive model complexity
 - Predictions driven by unstable variables

The solution is not simply to gather a high training score. Performance must be evaluated on data the model did not use to learn.

12. Analytical traps

Data leakage

- Data leakage occurs when information unavailable at prediction time enters model training.
- Always ask: “Would this information genuinely be available at the moment the prediction is made?”

Predicting whether a student will pass a course during the first month

Including the final-exam score would make the model look highly accurate, but the information would not exist when the prediction is needed.

12. Analytical traps

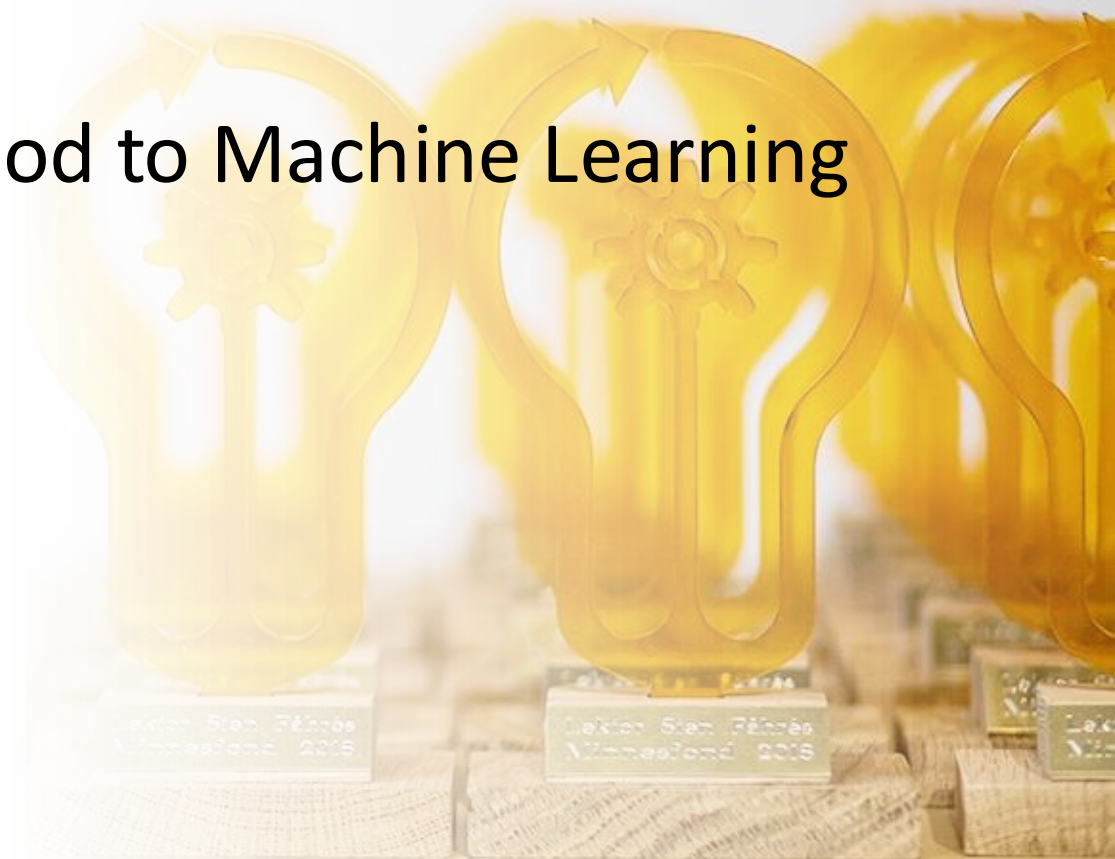
Statistical versus practical significance

- A relationship can be statistically significant but too small to matter in practice.
- Interpret findings using:
 - Effect size
 - Confidence intervals
 - Real-world costs and benefits
 - Study design
 - Domain knowledge
 - Consequences for affected groups

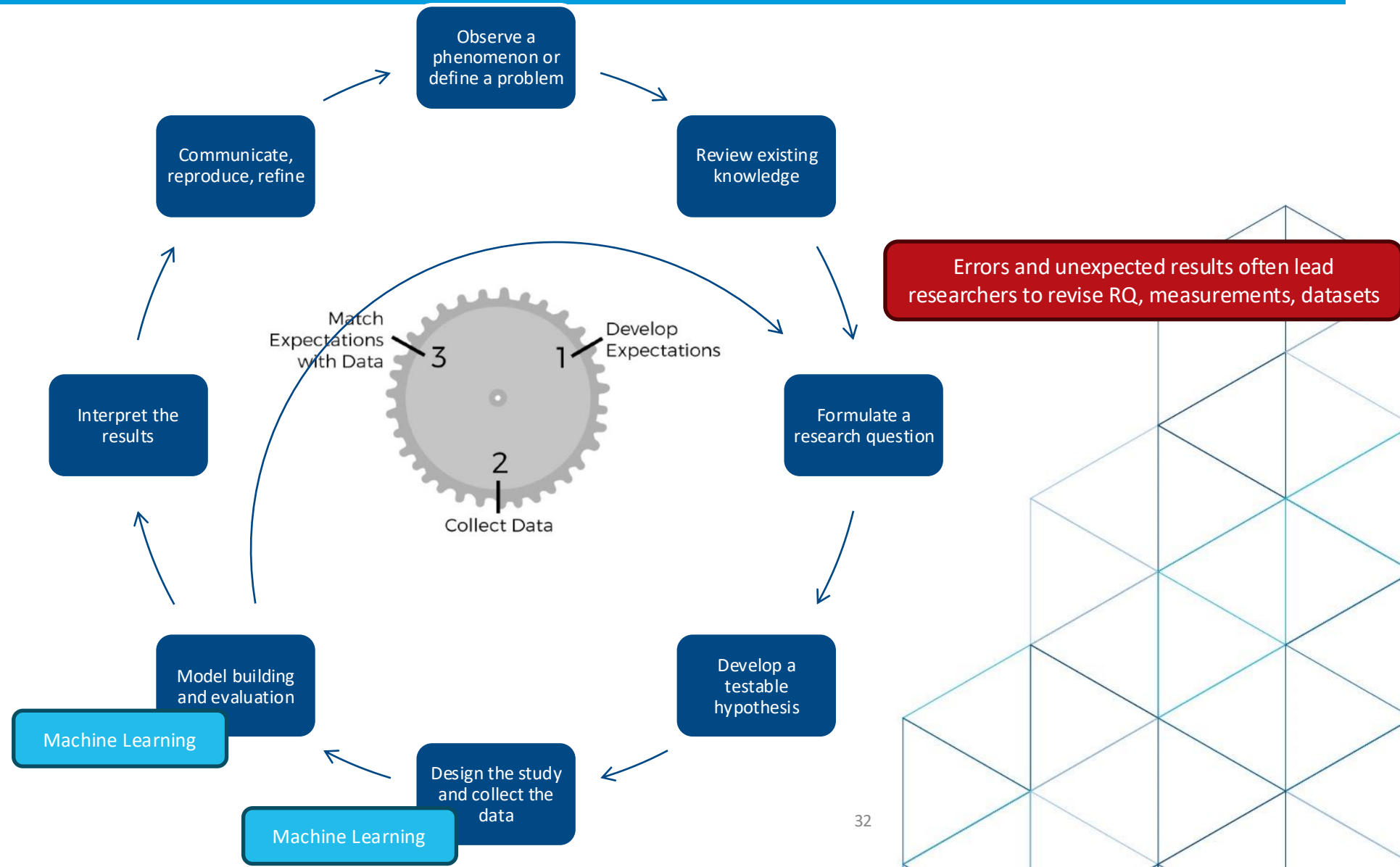
An additional hour of weekly study is associated with a 0.1-point increase on a 100-point grading scale

With a sufficiently large sample, this result might be statistically significant. However, the effect may have little educational importance.

Part V – From the Scientific Method to Machine Learning



13. ML: one tool within the scientific process



14. From a research question to a prediction task

The broad problem	A machine learning task
“Some students do not complete their courses”	“Using information available during the first four weeks, predict whether each student will complete the course”

This definition establishes:

- **Unit of analysis:** One student-course enrollment
- **Target:** Course completion
- **Features:** Early attendance, assignment submission, prior courses, and learning-platform activity
- **Prediction time:** End of week four
- **Target population:** Students enrolled in the relevant courses
- **Intended use:** Offering voluntary academic support

A useful model must be:

- Accurate on unseen data
- Evaluated against a meaningful baseline
- Free from data leakage
- Appropriate for the target population
- Examined for bias and unequal errors
- Useful for a clearly defined decision

15. Baseline and hypothesis

- A model should be compared with a meaningful baseline.
- Possible baselines include:
 - Predicting the most common outcome
 - Using a simple rule
 - Using logistic regression
 - Following the institution's current decision process

Two ways of setting the hypothesis:

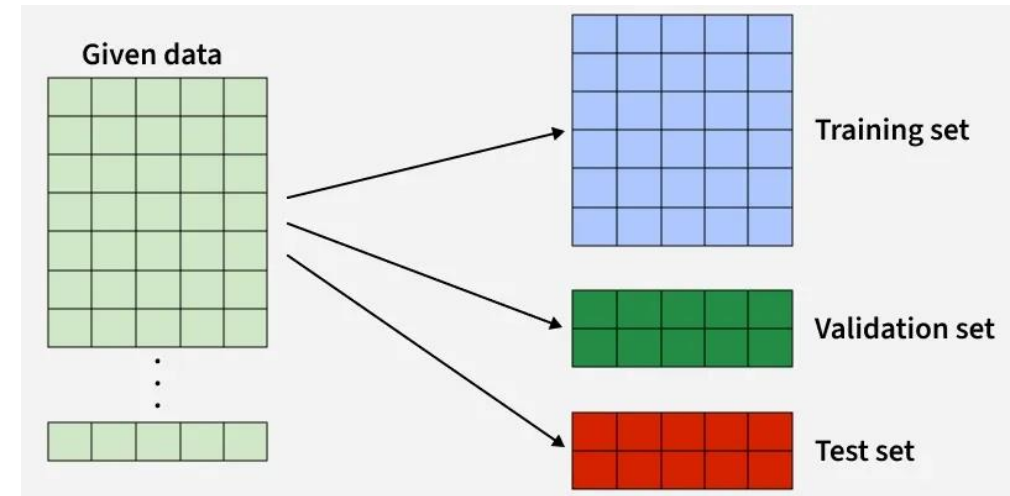
“Does the model achieve 85% accuracy?”

“Does the model improve on a relevant baseline in a way that supports the intended decision?”

16. Training, validation, and testing

In machine learning, A dataset is commonly divided into three parts:

- **Training set:** Used to fit the model
 - The algorithm learns patterns and estimates its parameters from these examples.
- **Validation set:** Used to make development choices
 - Compare models, select features, tune settings and choose decision thresholds.
- **Test set:** Used for the final evaluation
 - Estimate how the completed model performs on previously unseen data.
- **Illustrative split:** 70% training · 15% validation · 15% testing



The test set must not influence model development

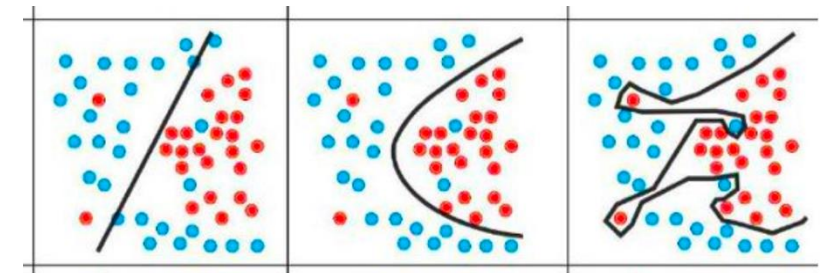
Repeatedly checking test performance and modifying the model turns the test set into another validation set.

17. Evaluation requires more than accuracy

- Accuracy can hide important errors.
- The appropriate metric depends on the consequences of different mistakes
 - A false negative: failing to identify a student who needs support.
 - A false positive: offering support to a student who may not need it.What is the cost of each error?
- Evaluation requires more than one score
 - Compare against a **simple baseline**
 - Evaluate using **unseen data**
 - Choose metrics that reflect the real objective
 - Consider the consequences of different errors
 - Check performance across relevant groups and conditions

18. Generalization and changing conditions

- **Underfitting:** The model is too simple to capture important patterns.
 - Poor performance on training data
 - Poor performance on unseen data
- **Good generalization:** The model captures useful patterns that also apply to new cases.
 - Good training performance
 - Similar performance on unseen data
- **Overfitting:** The model learns noise or details specific to the training data.
 - Excellent training performance
 - Much worse performance on unseen data



A useful model must generalize, not memorize

Part VI - Consolidation



26. Your journey through the course

- **Final Seminar:** Present a data science application:
 - Select an interesting application
 - Think about different aspects of such application
 - Connect your application to the content of each lecture
 - Work in groups!

You are going to get a series of assignments that will help you to prepare this task incrementally through the course

26. Your journey through the course

- For this week:
 1. For your data science task, discuss how it can be solved by considering qualitative as well as quantitative research methods.
 2. Reflect on data needed for each method.
 3. Develop a research question for each method.
 4. Discuss the possible outcomes/conclusions for each method.
 5. Submit the report in Blackboard

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